॥ श्रीगणेशाय नमः ॥

Averages

Properties of averages

Consider n elements $a_1, \ldots, a_n$.

$$\text{Avg} = \frac{a_1 + a_2 + \cdots + a_n}{n} = A$$

- Adding K to each term:

$$\text{Avg} = \frac{(a_1+K) + (a_2+K) + \cdots + (a_n+K)}{n} = \frac{a_1 + \cdots + a_n + nK}{n} = A + K$$

$$A' = A + K$$

- multiplying by K,

$$A' = \frac{a_1K + a_2K + \cdots + a_nK}{n} = KA \qquad A' = KA$$

- $a_1, a_2, \ldots, a_k, a_{k+1}, \ldots, a_n$

$\underbrace{a_1, a_2, \ldots, a_k}_{\text{avg} = A_k}$ $\underbrace{a_{k+1}, \ldots, a_n}_{\text{avg} = A_{n-k}}$

$$A = \frac{kA_k + (n-k)A_{n-k}}{n}$$

$$A = \frac{kA_k + (n-k)A_{n-k}}{k + (n-k)} \quad \text{(weighted avg.)}$$

If given group of n elements has avg A_n and another group of m elements has avg A_m, the overall avg is

$$A = \frac{nA_n + mA_m}{n+m}$$

- Adding a new element to a group of n elements:

$\underbrace{a_1, \ldots, a_n}$

$$\text{Avg} = \frac{a_1 + \cdots + a_n}{n} = A$$

case 1: $a_{n+1} > A$

$$A' = A + \frac{(a_{n+1} - A)}{(n+1)}$$



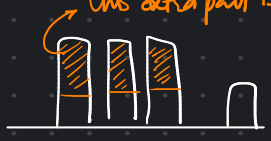
distributed among $(n+1)$ each gets $\frac{a_{n+1} - A}{(n+1)}$

so avg $= A + \frac{a_{n+1} - A}{(n+1)}$

Diff. is evenly distributed among $(n+1)$ elements.

case 2: $a_{n+1} < A$

$$A' = A + \frac{(a_{n+1} - A)}{(n+1)}$$



$$a_{n+1} + n\left(\frac{A - a_{n+1}}{(n+1)}\right)$$

- Removing an element from group:

case: If element < avg, then $A' < A$.

10 20 30 40 50

Avg = 30

Remove 50

Avg (10, 20, 30, 40) = 25 < 30

$A' < A$

case: If elem > avg:

$\text{avg}' > \text{avg}$

- Replacing an element in the group:

  If replaced elem $>$ elem $\longrightarrow$ $A' > A$

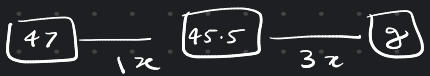
  $<$ $<$

- Avg of $1, 2, 3, \ldots, n = \frac{(n+1)}{2}$
- Avg of $1^2, 2^2, \ldots, n^2 = \frac{(n+1)(2n+1)}{6}$
- Avg of $1^3, 2^3, \ldots, n^3 = \frac{(n)(n+1)^2}{4} = \frac{\left(\frac{n(n+1)}{2}\right)^2}{n}$
- Avg of $2, 4, 6, 8, \ldots, 2n = \frac{2(1+2+\cdots+n)}{n} = (n+1)$
- Avg of $1, 3, 5, 7, \ldots, 2n+1 = n$
- Avg of consecutive #s $a, b, c, \ldots, n = \frac{a+n}{2}$
- Avg of first n multiples of K, i.e. $K, 2K, 3K, \ldots, nK = \frac{K(n+1)}{2}$

Q. Ratio of boys to girls in a class of 48 is 3:1. Avg weight of boys is 47, that of the whole class is 45.5. What's the avg wt of girls?

→ $\frac{3(47)+1g}{3+1} = 45.5 \quad \Rightarrow g = 41$

$b:g = 3:1$



$x = 1.5$

$3x = 4.5$ $\quad 45.5 - 4.5 = 41$

Q. A person buys 68 kg apples at rs. 63/kg, 85 kg apples at rs. 81/kg. Avg. price?

→ 68kg @₹63/kg, 85kg @₹81/kg $\quad 68:85 = 4:5$

so ₹ $x-63$: ₹ $81-x$

$= 5:4$

$\frac{x-63}{81-x} = \frac{5}{4} \Rightarrow 4x - 63(4) = 81(5) - 5x$

$\Rightarrow 9x = 657$

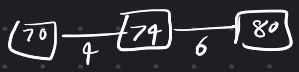
$x = 73$

Q. 27 boys have avg score = 70. 18 girls have avg score = 80. Avg of class?

→ $B:g = 27:18 = 3:2$



$\frac{2}{5} \times 10 = 4 \quad \frac{3}{5} \times 10 = 6$



Q. Avg of 3 #s is 143. The 1st and 3rd #s are in the ratio 7:9. The 2nd # is 40% more than the 3rd #. Find the sum of the 2nd and the 3rd #.

→ $a+b+c = 143 \cdot 3 = 429$

$7x + \frac{140}{100}(9x) + 9x = 429$

$(700 + 1260 + 900)x = 42900$

$x = \frac{42900}{2860} = 15$

Sum of 2nd & 3rd #s $= \frac{140}{100}(9x) + 9x = 9x\left(\frac{140}{100} + \frac{100}{100}\right) = 9(15)\left(\frac{240}{100}\right) = 324$

Q. Avg age of a, b, c, d, e is 40.2. Avg age of a and b = 34.5. Avg age of c and d = 41.5. Age of e?

→ $a+b+c+d+e = 40.2 \times 5 = 201 \quad a+b = 69 \quad c+d = 83$

$e = 201 - 69 - 83 = 49$

Q. Avg age of 100 employees is 32 years. 2/5th of them are women. The ratio of average ages of men to women is 6:7. What's the avg age of men?

→ #m : #w = 3:2

$32 = \frac{6x(3) + 7x(2)}{3+2}$

$\Rightarrow x = \frac{160}{32} = 5$

$6x = 30$

Q. In Ashish's opinion, his weight is >55kgs but <62kgs. His father doesn't agree. He thinks that Ashish weighs > 50 but < 60kgs. His sister's view is that his weight can't be > 58kgs. If all of them are correct, what is the avg of diff prob weights of Ashish?
a. 57.5 b. 56 c. 68 d. Data inadequate e. None of these

→ $x \in (55, 62)$ $\quad x \in (50, 60)$ $\quad x \leq 58$

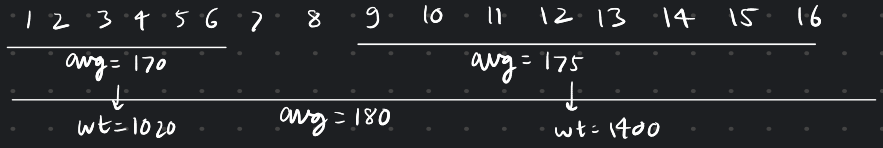
56, 57, 58, 59, 60, 61 $\quad$ 51, 52, 53, ..., 60 $\quad$ 1, 2, ..., 58

common: 56, 57, 58

Avg = 57

Q. Avg height of 1st 6 students is 170cm. The avg height of last 8 students is 175cm. The avg height of total 16 students is 180cm. Find avg height of rest 2 students.

→



$$180 = \frac{1020 + 1400 + (\text{wt of 2 students})}{16}$$

$w = 960$

Avg = 230 cm

Q. Avg wt of p, q, r, s, t is x+6 kg. That of r and t is x-6 kg. If u is added, avg of p, q, r, s, t, u is reduced by 5 kgs. Find the value of x if the avg wt of p, q, s, u is 94.5 kgs.

→ $p+q+s+u = 378$ $\quad r+t = 2x-12$ $\quad p+q+r+s+t = 5x+30$ $\quad p+q+r+s+t+u = (x+1)6$

$p+q+s = 5x+30-2x+12$

$= 3x+42$

$u = 378 - 3x - 42$

$5x+30+378-3x-42 = (x+1)6 = 6x+6$

$4x = 378+30-42-6$

$x = 90$

Q. Avg age of a family of 6 members 4 years ago was 25 yrs. Meanwhile, a child was born. The avg is still the same. Find the present age of the child.

→ Say all 6 members are aged x.

$\frac{6(x-4)}{6} = 25$

$x = 29$ ← curr. age

say child was born y yrs ago

$\frac{(29)(6)+y}{7} = 25$

$y = 1$

---

4 yrs ago:

$A_6 = 25$

$S_6 = 150$

∴ Today:

$S_6 = 150 + 4(6) = 174$

Now say age of child is x

$\frac{174+x}{7} = 25$

$x = 1$

Q. Avg of 4 consecutive natural #s is 67.5. One of them is div by 4. Find avg of all natural #s below this #.

→ $a+b+c+d = 67.5 \times 4 = 270$

$(n)+(n+1)+(n+2)+(n+3) = 270$

$4n = 264$

$n = 66$

66, 67, 68, 69

↑ div. by 4

$\text{avg}(1+2+3+\cdots+67) = \frac{67+1}{2} = 34$